Introduction

A country's road infrastructure is an essential part of its transportation network and is essential to daily transportation and economic activity. Road networks in Sri Lanka are constantly subjected to high traffic volumes, adverse weather, and deteriorating building materials, all of which cause different types of road surface damages. Potholes, cracks, raveling and rutting are common road surface problems that have an adverse impact on driving comfort, increase vehicle operating costs and place road user's safety at danger. Effective road maintenance and management thus depend on timely identification and evaluation of such surface damage.

Currently, many road condition assessments are carried out by skilled workers using manual visual inspections. Despite its wide acceptance, this approach is expensive, labor-intensive, and time-consuming, particularly when monitoring large road networks. Furthermore, manual inspections are subjective in nature and can produce inconsistent results based on the inspector's experience and the conditions of the environment at the time of the inspection. These drawbacks emphasis the importance of an automated and trustworthy method for identifying and assessing road surface damage.

Automated road condition monitoring systems are becoming more possible due to the rapid growth of image and video processing technologies. Road surfaces can be continuously captured by video-based image processing systems, which can then analyze individual frames to identify damaged areas. Road surface details can be more clearly seen by using the right picture enhancement techniques to reduce problems like low contrast, noise and uneven illumination. Additionally, damaged areas can be isolated and extracted using image segmentation techniques based on difference in edge information, texture, and intensity. These methods offer a scalable and affordable alternative for detecting road damage, especially when using commonly available equipment like cell phone cameras.

The objective of this project is to create and deploy a video-based image processing system that uses video footage taken from specific road segments to identify, segment, and evaluate road surface damages. The suggested system focusses on detecting various road surface deteriorations, such as potholes and various types of cracking, and computationally evaluating the extent of the damages found. To ensure accurate and trustworthy detection of road surface damages under various environmental circumstances, the developed algorithm's performance is evaluated using both qualitative observations and quantitative measurements.